BMI Calculator for a Team

Week 1 Practice - Problem 3 | Java

Problem Summary

Use height and weight arrays to calculate BMI = weight / (height * height), classify it using the source thresholds, and print a wellness table.

Java Solution

```
public class Main {
    static String getBmiStatus(double bmi) {
        if (bmi < 18.5) return "Underweight";
        if (bmi < 25.0) return "Normal";
        if (bmi < 30.0) return "Overweight";
        return "Obese";
    }
    static void printWellnessReport(double[] heights, double[] weights) {
        if (heights.length != weights.length) {
            System.out.println("Heights and weights must have equal lengths");
            return;
        }
        System.out.println("Person | Height (m) | Weight (kg) | BMI | Status");
        for (int i = 0; i < heights.length; i++) {
            double bmi = weights[i] / (heights[i] * heights[i]);
            System.out.printf("Person %d | %.2f | %.2f | %.2f | %s%n",
                i + 1, heights[i], weights[i], bmi, getBmiStatus(bmi));
        }
    }
    public static void main(String[] args) {
        printWellnessReport(new double[]{1.75, 1.60}, new double[]{70, 90});
    }
}
```

Sample Output

```
Person | Height (m) | Weight (kg) | BMI | Status
Person 1 | 1.75 | 70.00 | 22.86 | Normal
Person 2 | 1.60 | 90.00 | 35.16 | Obese
```

Complexity / Key Notes

Time: O(n) for n people. Extra space: O(1) beyond input arrays. BMI is a screening measure, not a stand-alone diagnosis.